

A Full-stack Simulator for Heterogeneous AI Training Clusters

Sumit Kumar, Arjun Temura, Naman Sharma, Ramanjeet Singh, Meet Dadhania, Praveen Tamma, Satananda Burla, Abed Mohammad Kamaluddin, Rinku Shah



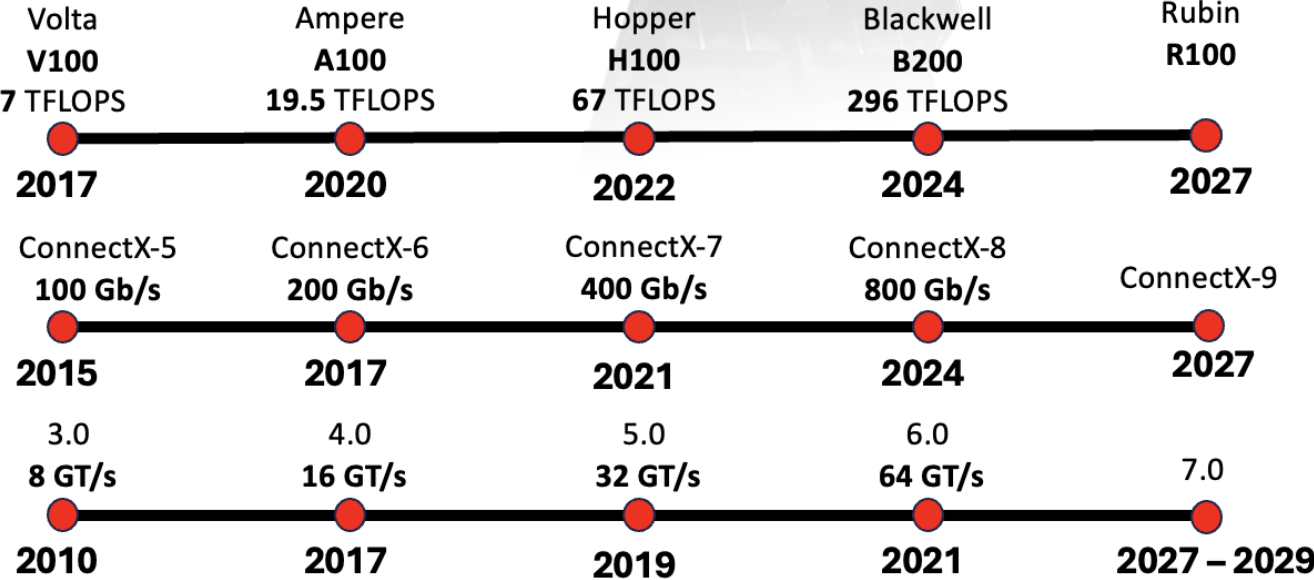
Evolution of the hardware



GPU architecture:
GPU type:
Compute power:

NIC :
NIC Bandwidth:

PCIe Express:
Transfer Rate
(per lane):



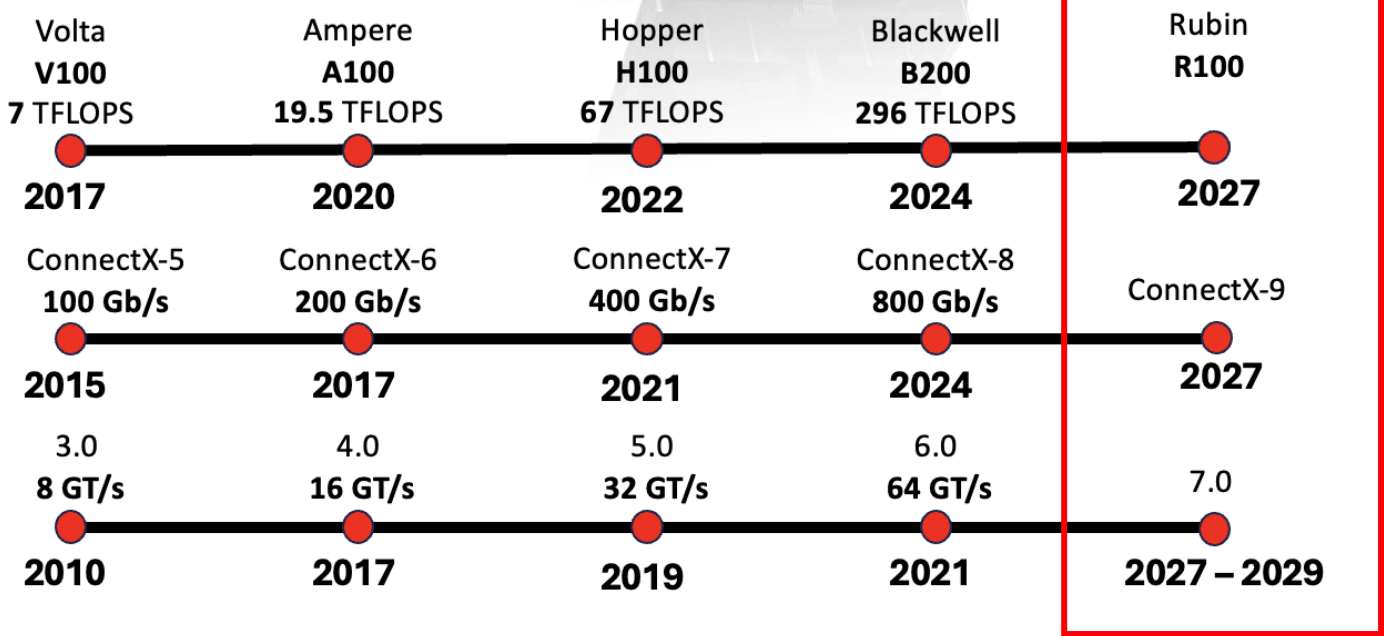
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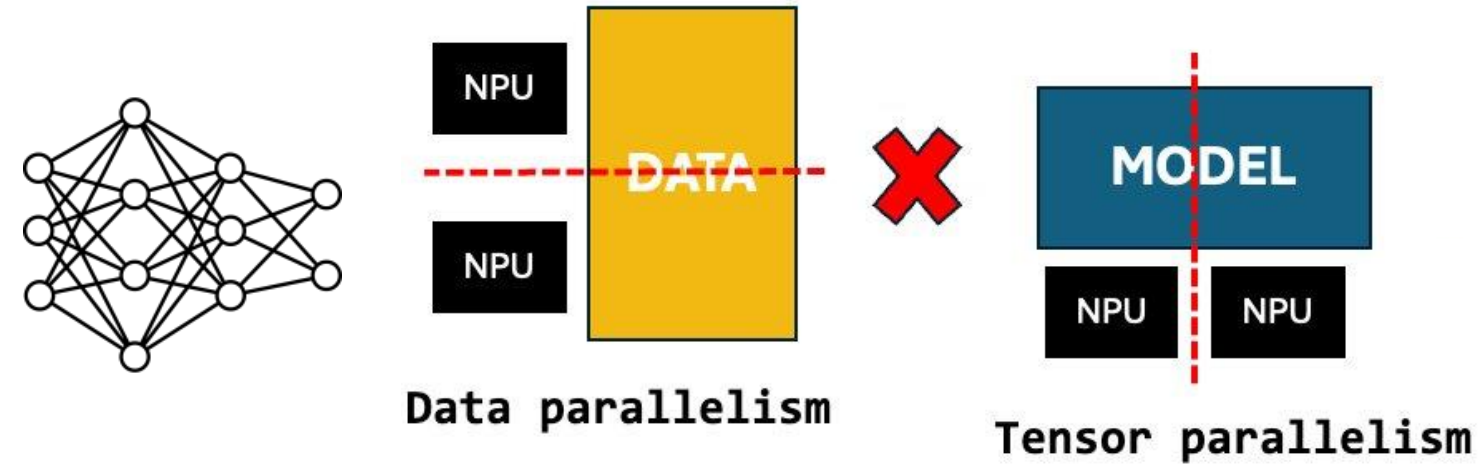
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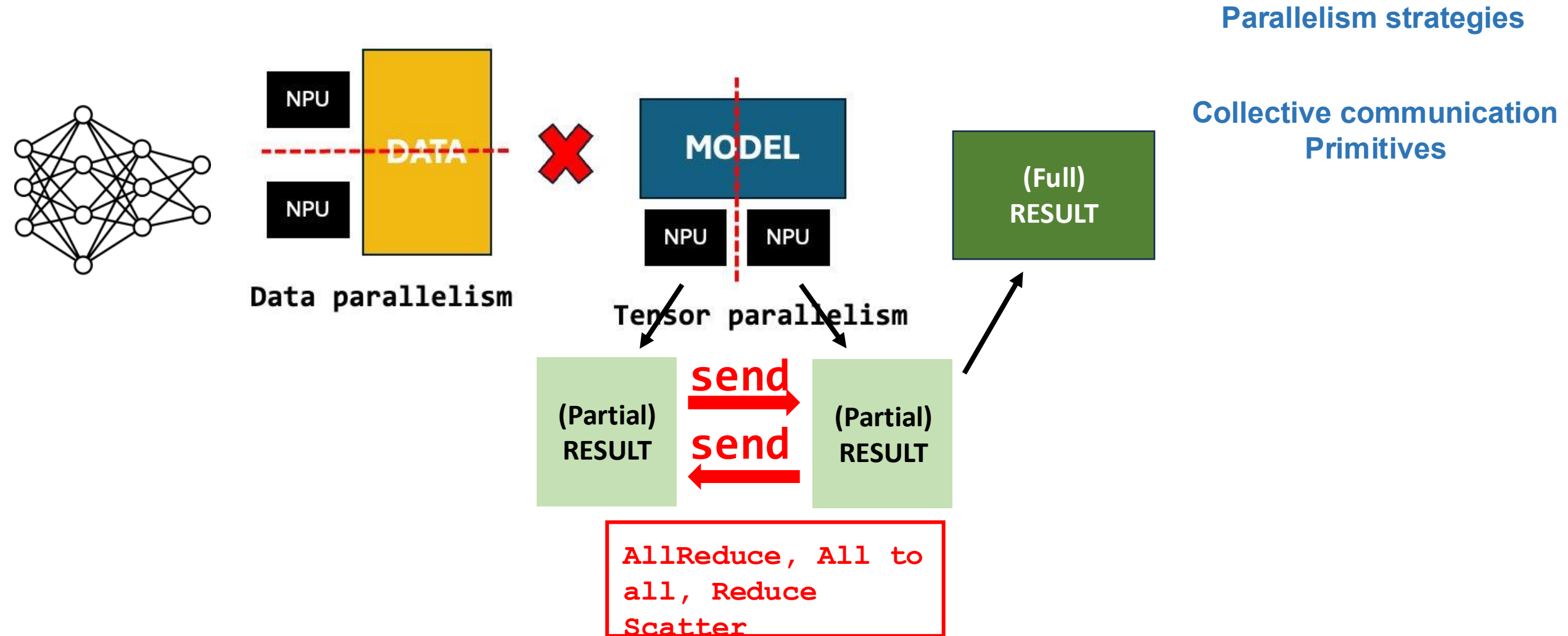
Simulation is necessary!

Complex AI training stack.

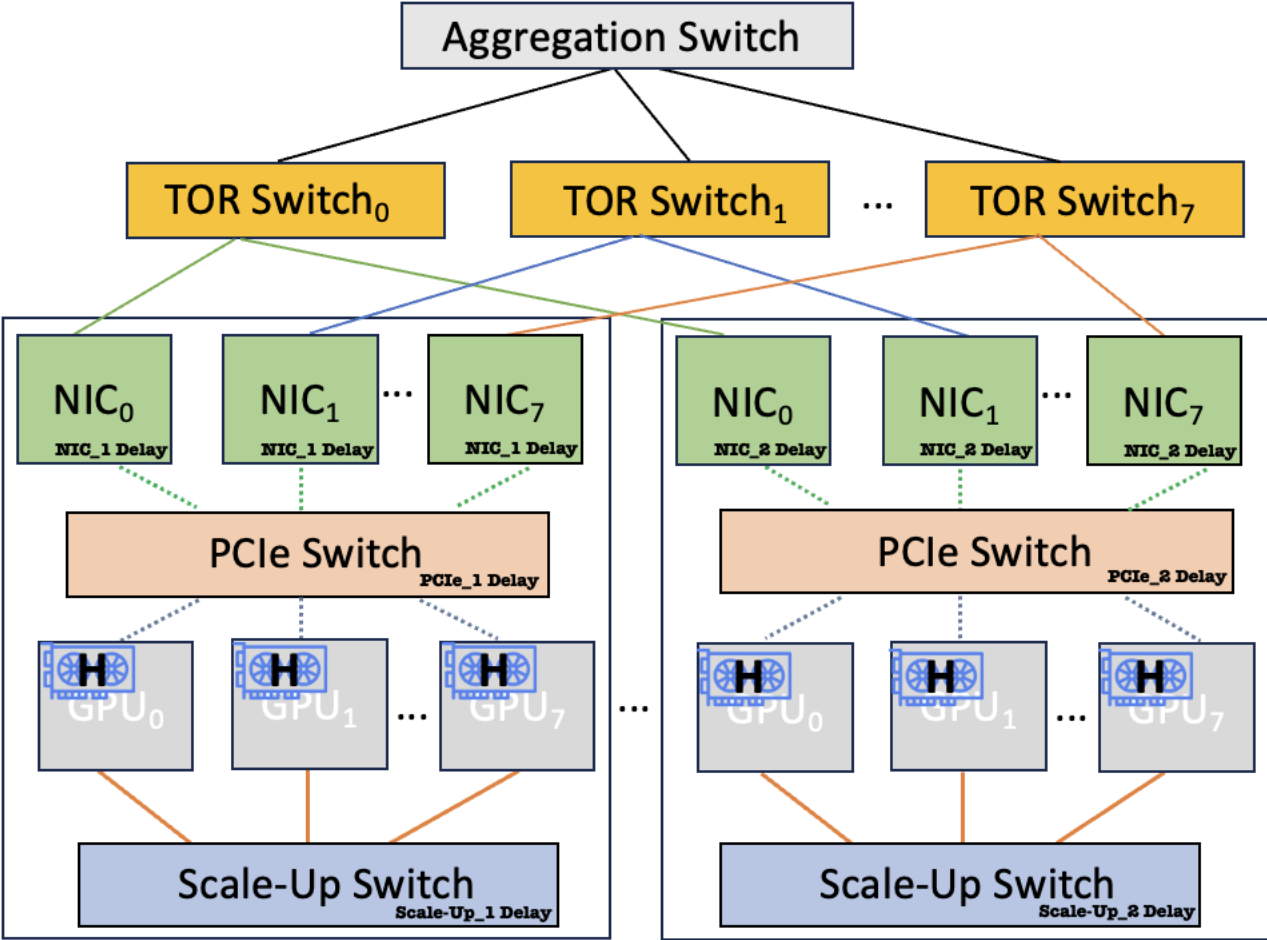
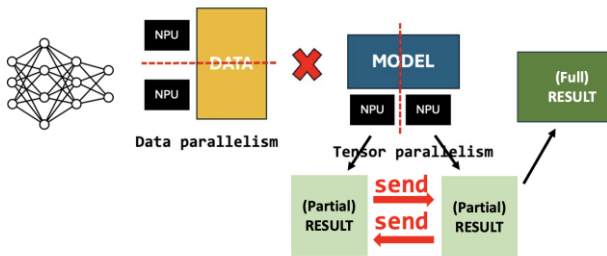
Parallelism strategies



Complex AI training stack.



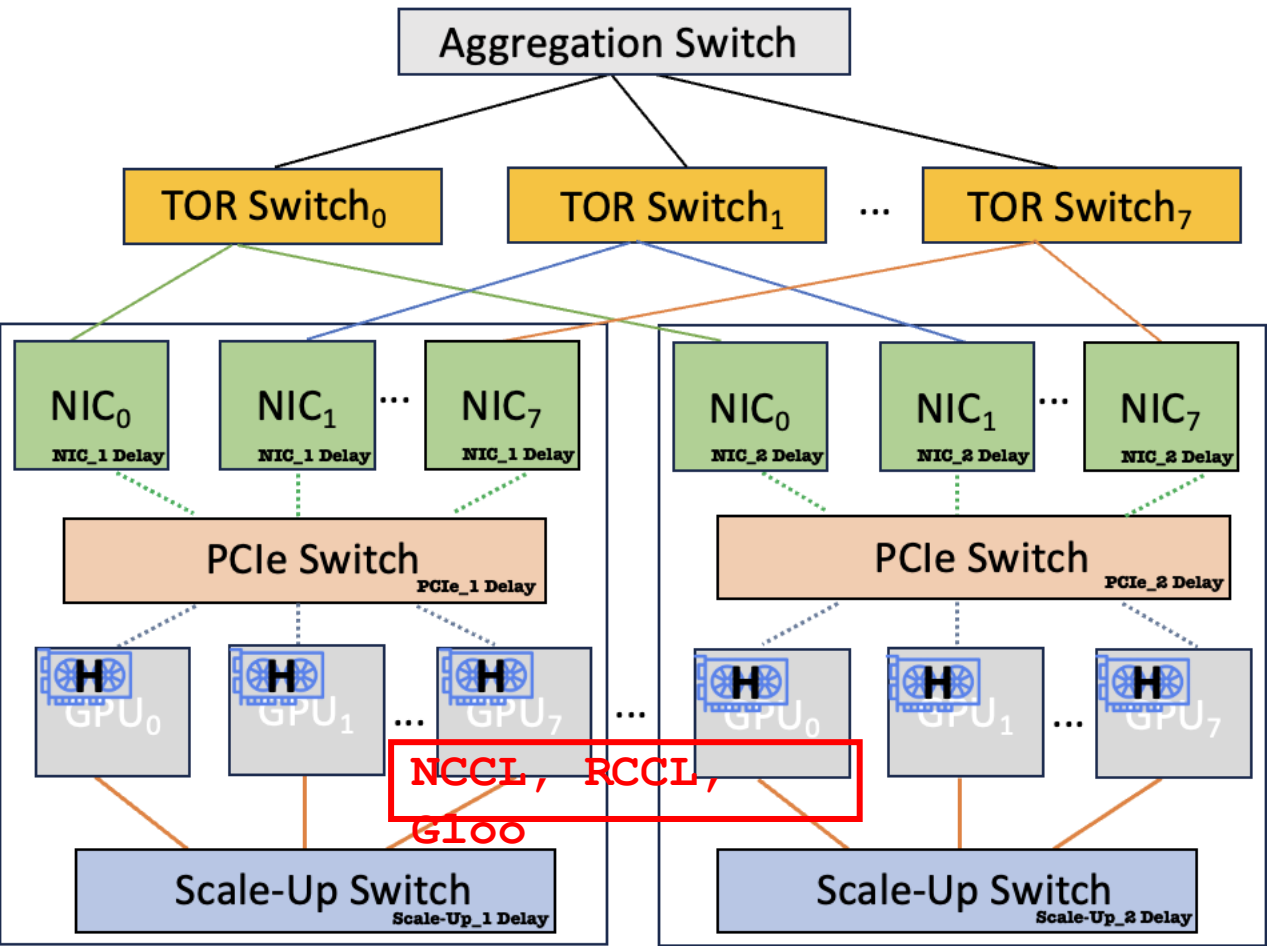
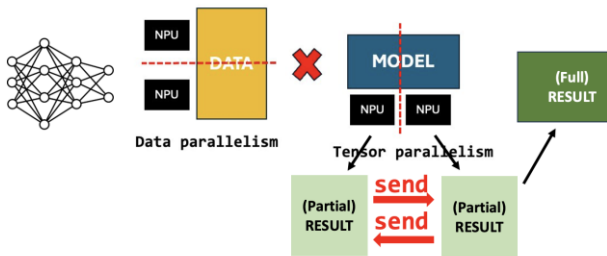
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Parallelism strategies

Collective communication
Primitives

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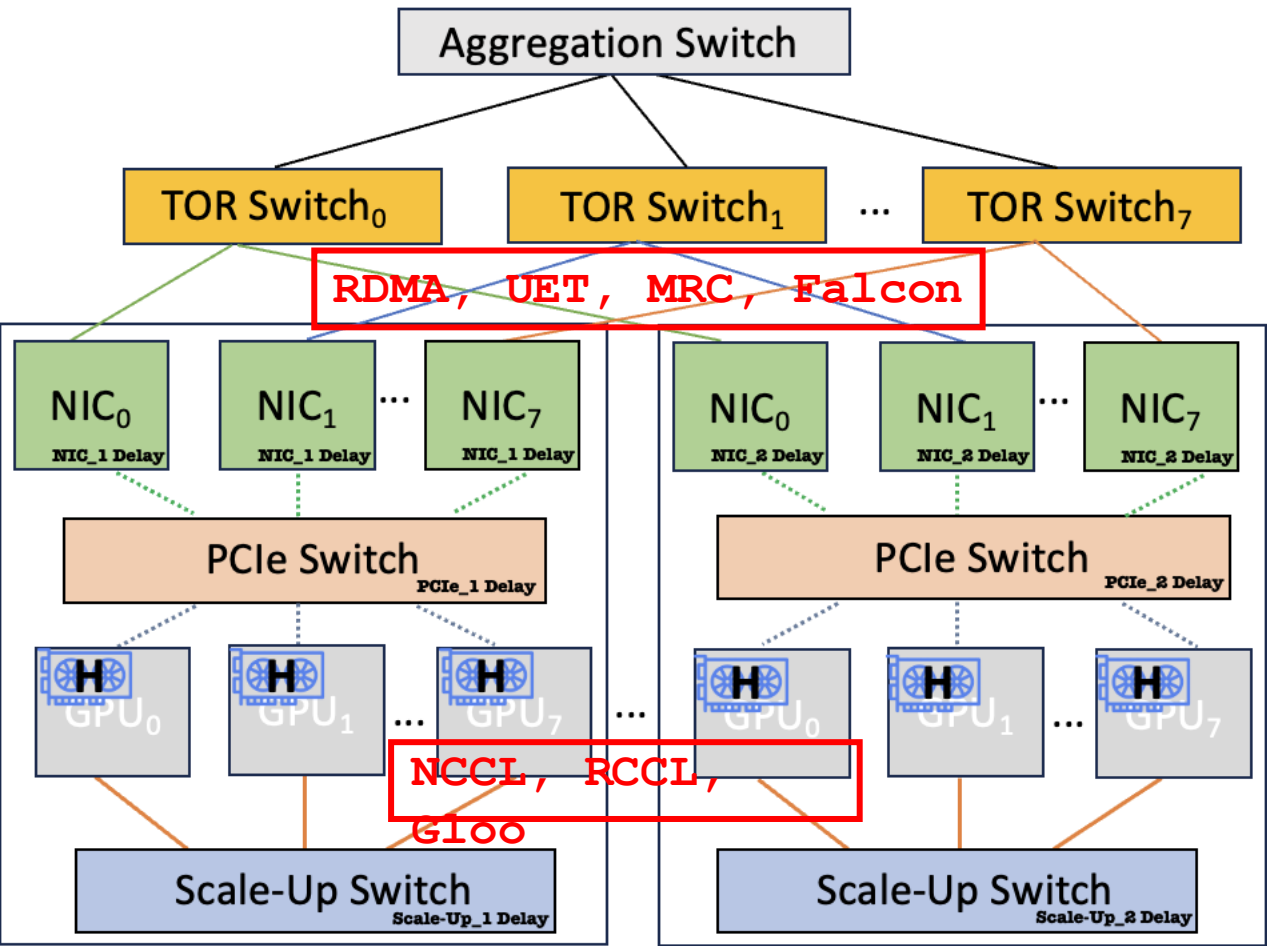
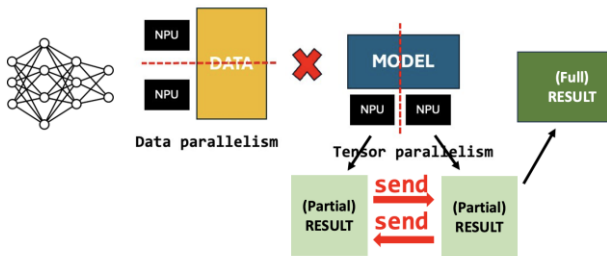


Parallelism strategies

Collective communication
Primitives

Collective communication
Library

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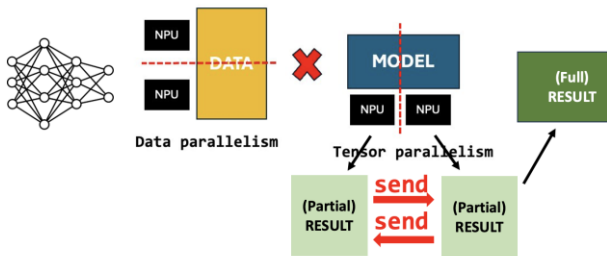
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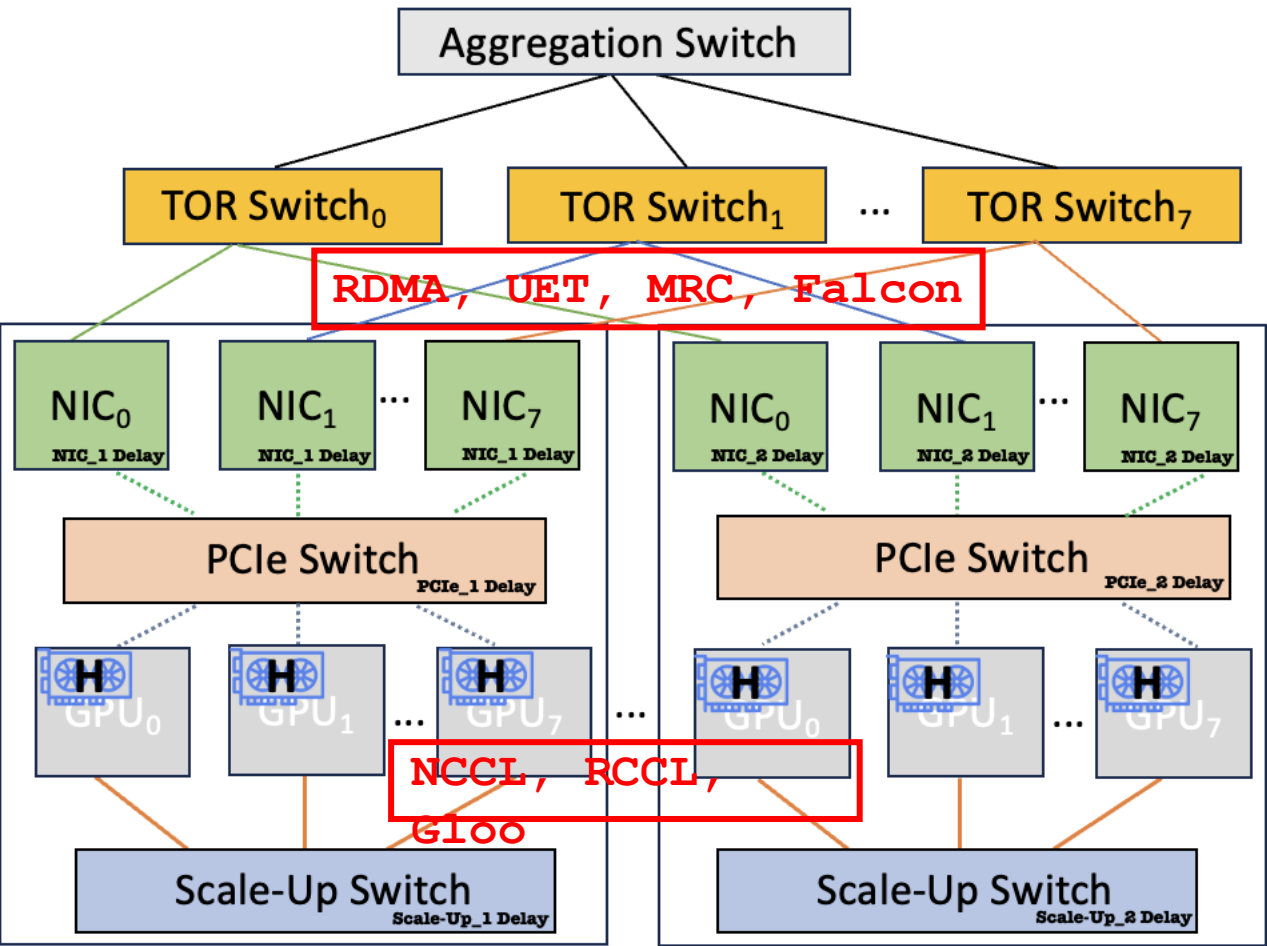
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Transport Protocol

Complex AI training stack.



GPUs, TPUs,
FPGAs, SmartNICs



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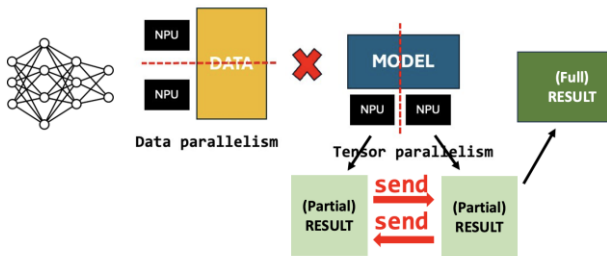
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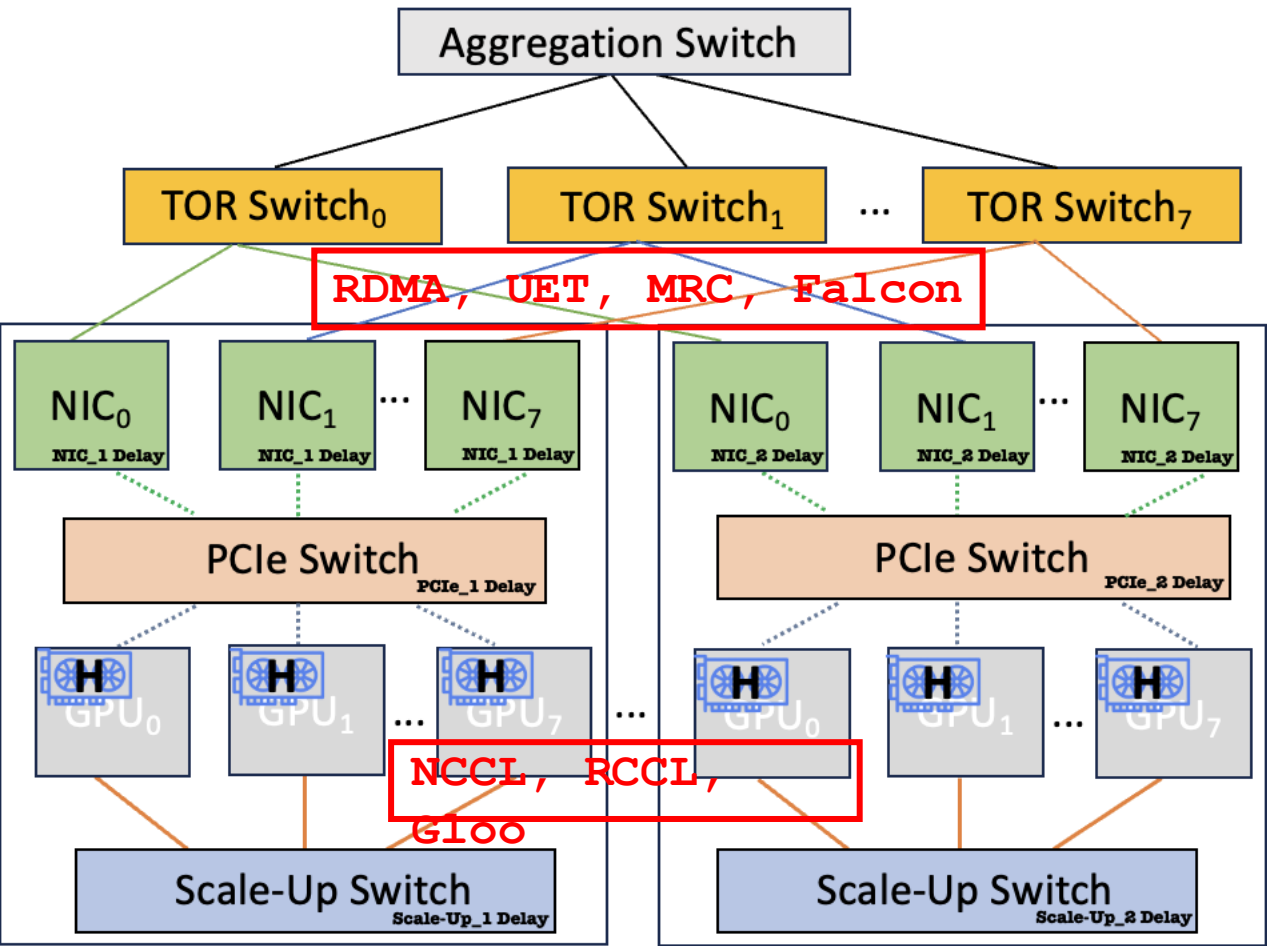
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Hardware configurations

Complex AI training stack.



GPUs, TPUs,
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Parallelism strategies

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Collective communication
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Transport Protocol

Hardware configurations

Distributed AI training simulation is necessary !

Existing simulation frameworks

Features	AstraSim ^[1]	Echo ^[2]	vTrain ^[3]	Multi-verse ^[4]	Atlahs ^[5]	SimAI ^[6]
Trace extrapolation support	×	✓	×	×	×	✓
Collective optimization support	×	✓	✓	×	✓	✓
Pipeline parallelism support (discrete-event)	×	×	×	×	✓	×
Network protocol simulation	✓	×	×	✓	✓	✓
Heterogeneous cluster simulation	×	×	×	×	×	×

[1] Astra-sim2.0: Modeling hierarchical networks and disaggregated systems for large-model training at scale

[2] Echo: Simulating distributed training at scale

[3] vtrain: A simulation framework for evaluating cost-effective and compute-optimal large language model training

[4] Accelerating Design Space Exploration for {LLM} Training Systems with Multi-experiment Parallel Simulation

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Why Heterogeneity Matters



V100, Volta
(‘17)

<



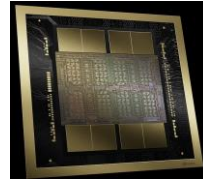
A100, Ampere
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H100, Hopper
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B100, Blackwell
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GPU and interconnect advances 3.0x and 1.4x annually!

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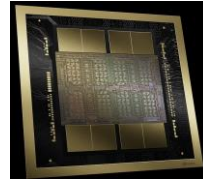
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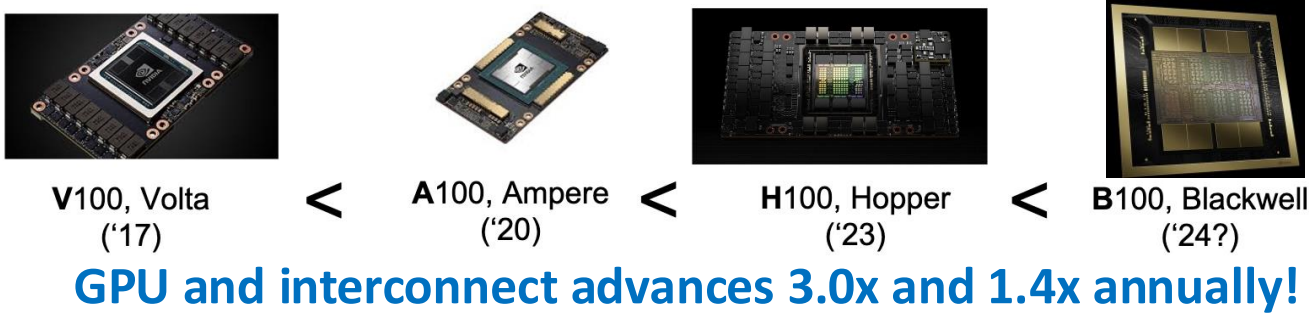
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**Modern AI clusters span multiple data centers
due to power demands in the tens of megawatts!**

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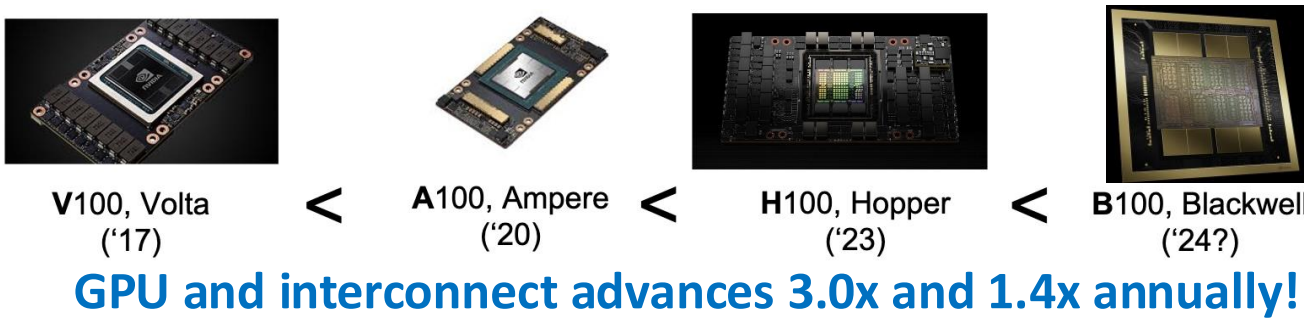


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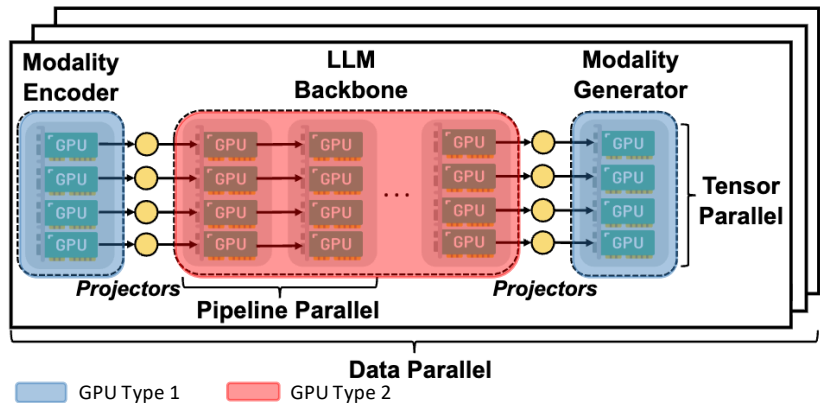
Hyperscale	Networking Stack	Compute
Google Cloud	GPUDirect RDMA, GPUDirect-TCPXO, GPUDirect-TCPX	A4X Max (GB300 Ultra superchips), A3 Ultra (NVIDIA H200 GPUs), A3 High (NVIDIA H100 GPUs).

Heterogeneous deployment of the model in cloud infrastructure!

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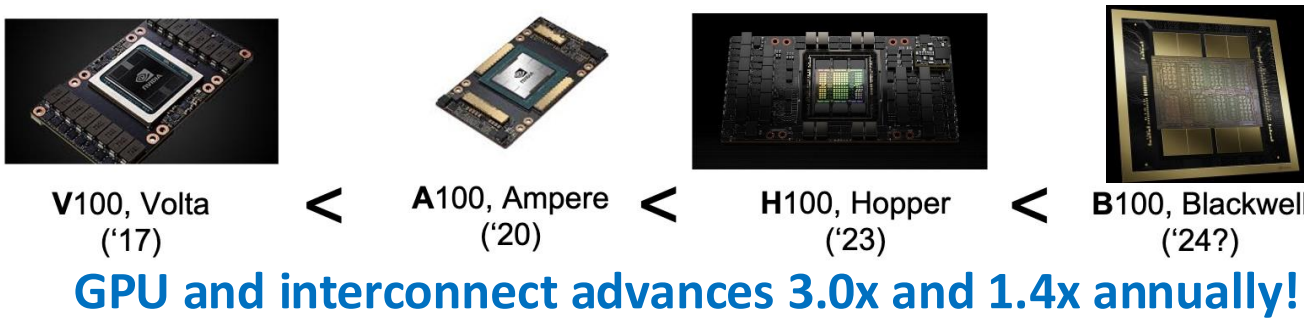


Leveraging device heterogeneity for improved utilization in multimodal training!

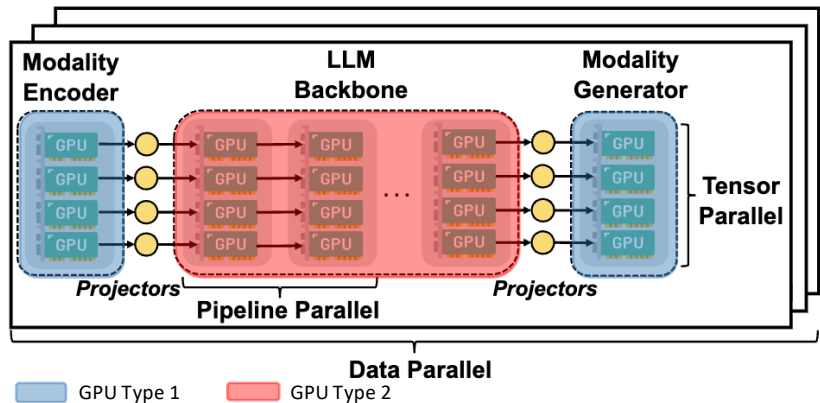
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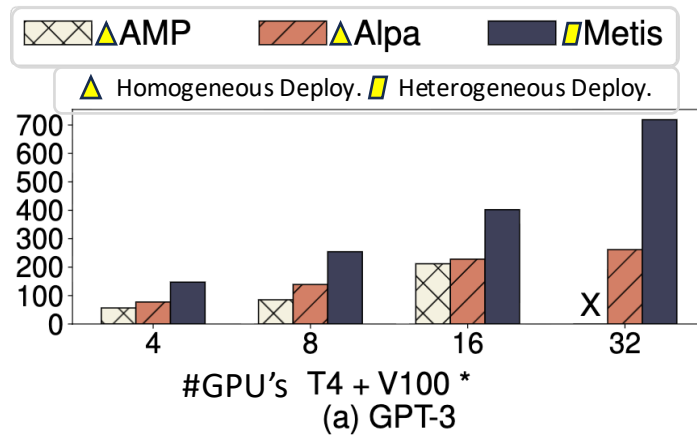
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Modern distributed training infrastructure is inherently heterogeneous !

Problems with heterogeneity

- Load Imbalance

Cluster is heterogeneous, but deployment is homogeneous

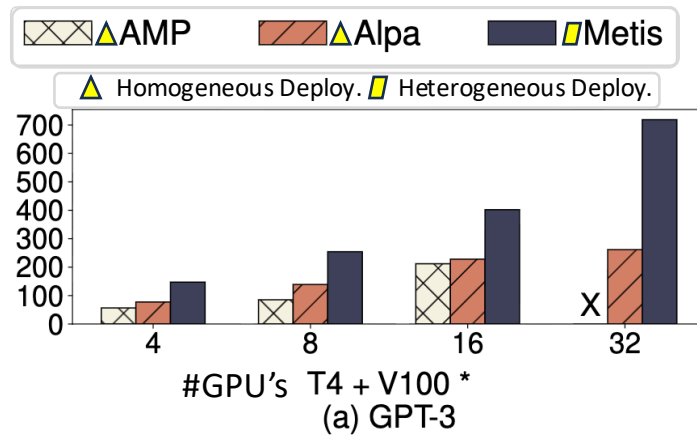


Metis achieves $\sim 2.7\times$ higher throughput than Alpa at 32 GPUs.

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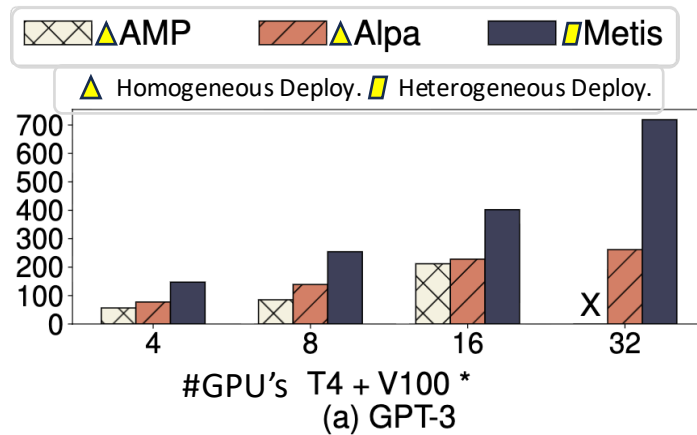
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Heterogeneous clusters require heterogeneity-aware deployment!

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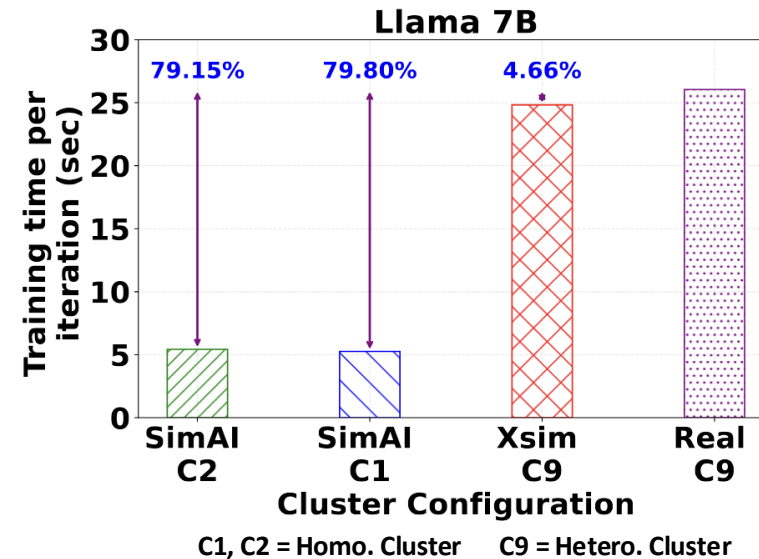
- Load Imbalance
- Naive collective operation

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SimAI → 80% prediction error

Xsim → 4.7% error

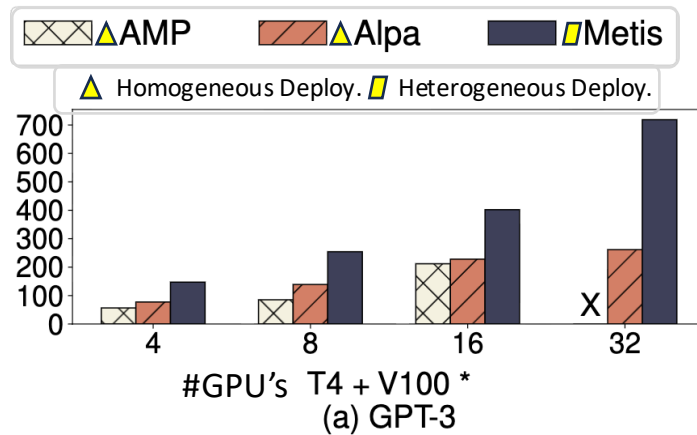
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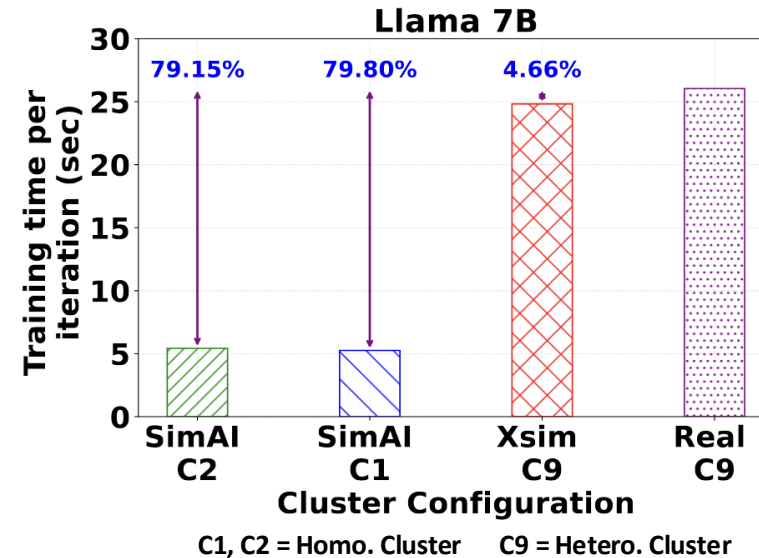
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Heterogeneous clusters require heterogeneity-aware deployment!

Heterogeneity-aware collectives are essential!

Heterogeneity-aware deployment related work

	Non-Uniform deployment strategies	Validate using Real world	Validate using Analytical Simulation	Full Stack Simulation	Heterogeneous Communication
HeterMoe^[1]	✓	✓	✗	✗	Send/Recv
Whale^[2]	✓	✓	✗	✗	Bridge
Metis^[3]	✓	✓	✗	✗	Centralized Resharding
Hexiscale^[4]	✓	✓	✓	✗	Leader
Hetauto^[5]	✓	✓	✗	✗	GCD-based Gather Scatter
AlpaComm^[6]	✓	✓	✗	✗	Cutpoint-Union

[1] Hetermoe: Efficient training of mixture-of-experts models on heterogeneous gpus

[2] Whale: Efficient giant model training over heterogeneous {GPUs}

[3] Metis: Fast automatic distributed training on heterogeneous {GPUs}

[4] HexiScale: Accommodating Large Language Model Training over Heterogeneous Environment

[5] HetAuto: Cross-Cluster Auto-Parallelism for Heterogeneous Distributed Training.

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Problem statement

Design a full-stack AI training simulator that predicts performance on AI clusters with heterogeneous compute nodes and interconnects

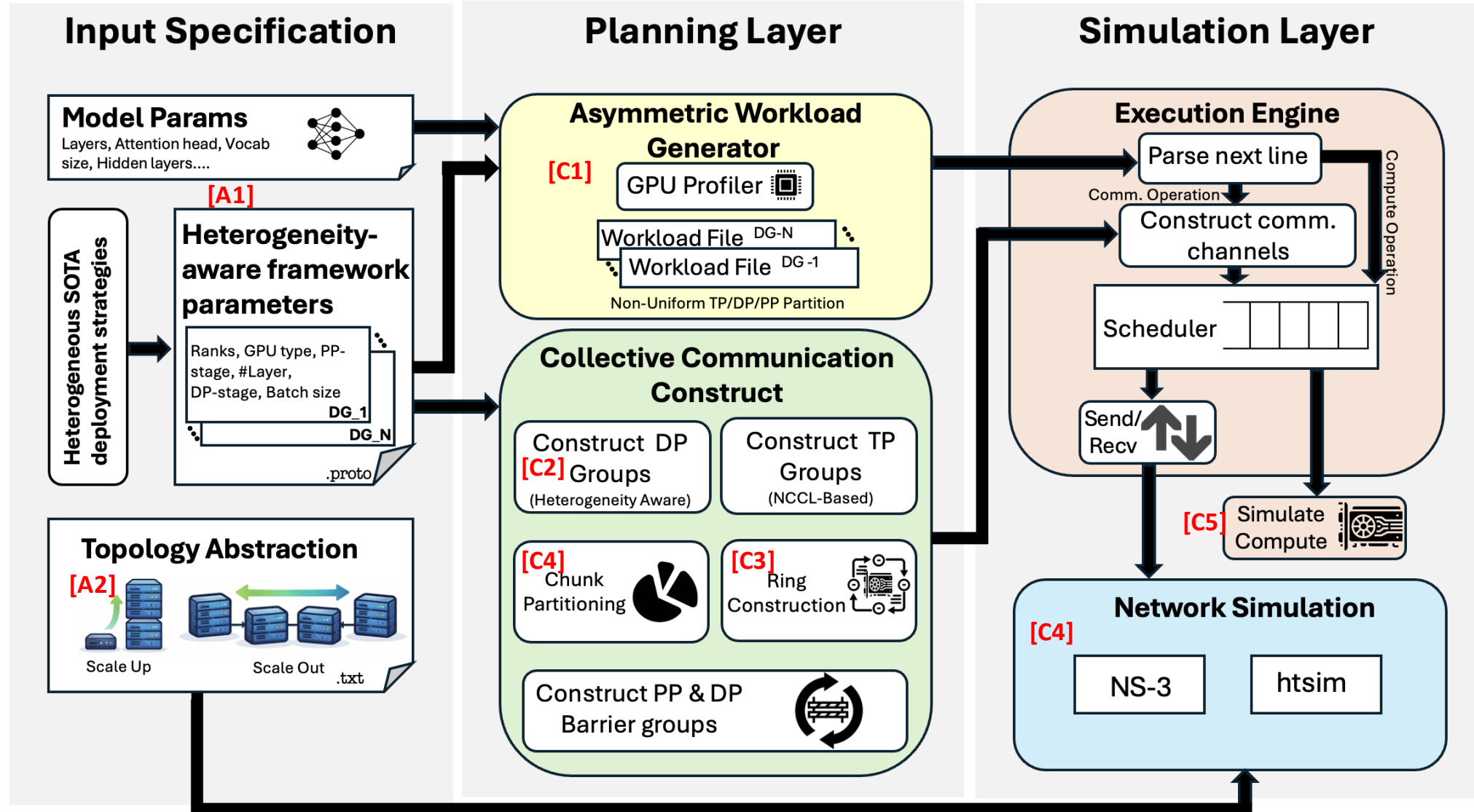
Contributions:

1. For heterogeneous AI training simulation, we identified
 - a) Requirements
 - b) Abstraction
2. Design primitives for
 - a) Collective communication
 - b) Tensor resharding
3. Design and implemented heterogeneous collective construct
 - a) Sweep line algorithm
 - b) LCM-based chunk partitioning
4. Reusable primitives for future PP strategies
5. Provide support for scale-up/scale-out networks
 - a) NS-3
 - b) htsim

Requirements

Objectives	Requirements
Non-Uniform workload partitioning	[A1] Device Group and Model Specification [A2] Heterogeneous Cluster and Host Topology Specification [C1] Asymmetric Workload Generation
Heterogeneity aware collective communication	[C2] Heterogeneity-aware Ring Construction [C3] Heterogeneity-aware Chunk Partitioning
Modelling heterogeneity-aware compute and interconnect	[C4] Scalable Network Simulation [C5] Heterogeneous Compute Simulation

Design



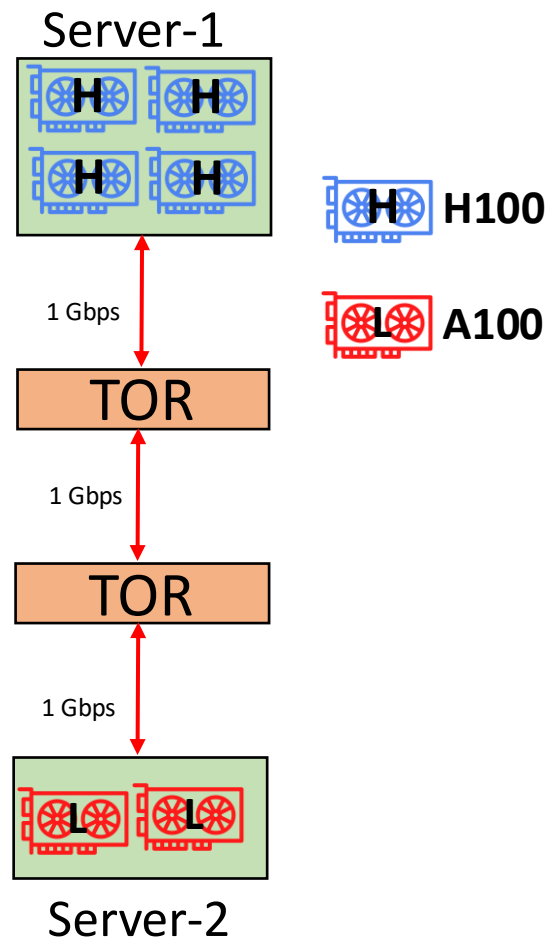
Evaluation questions

We evaluated Xsim on various parameters:

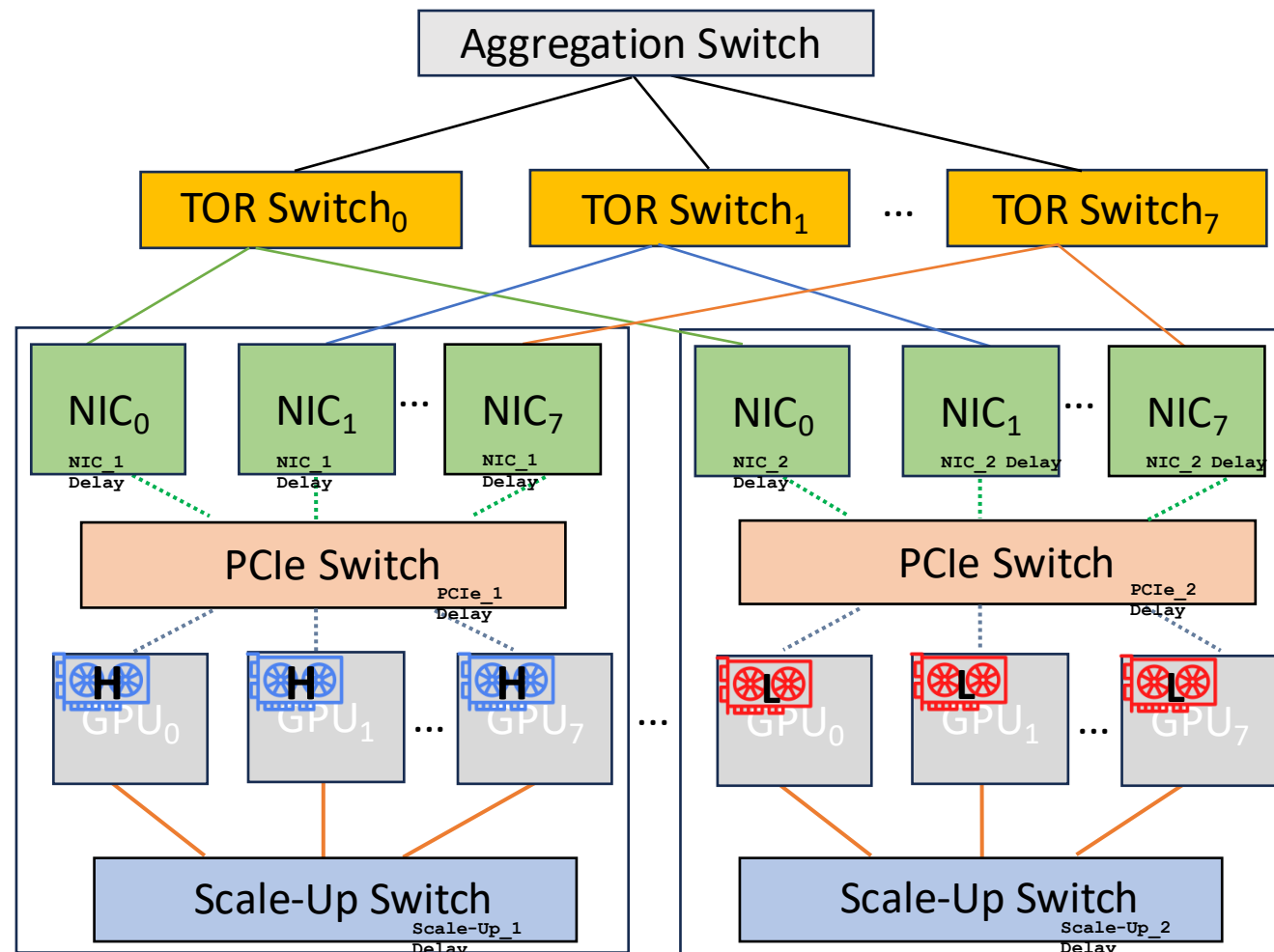
1. Accuracy:
 - Comparison with heterogeneous deployment with ground truth and analytical simulator.
 - Prediction accuracy on DP multi-ring Collective communication synchronisation.
2. SOTA Resharding Comparison
 - Comparison with SOTA Resharding solution.
3. Scalability:
 - htsim vs. ns-3 simulation runtime trade-offs

Setup

Real Cluster Setup

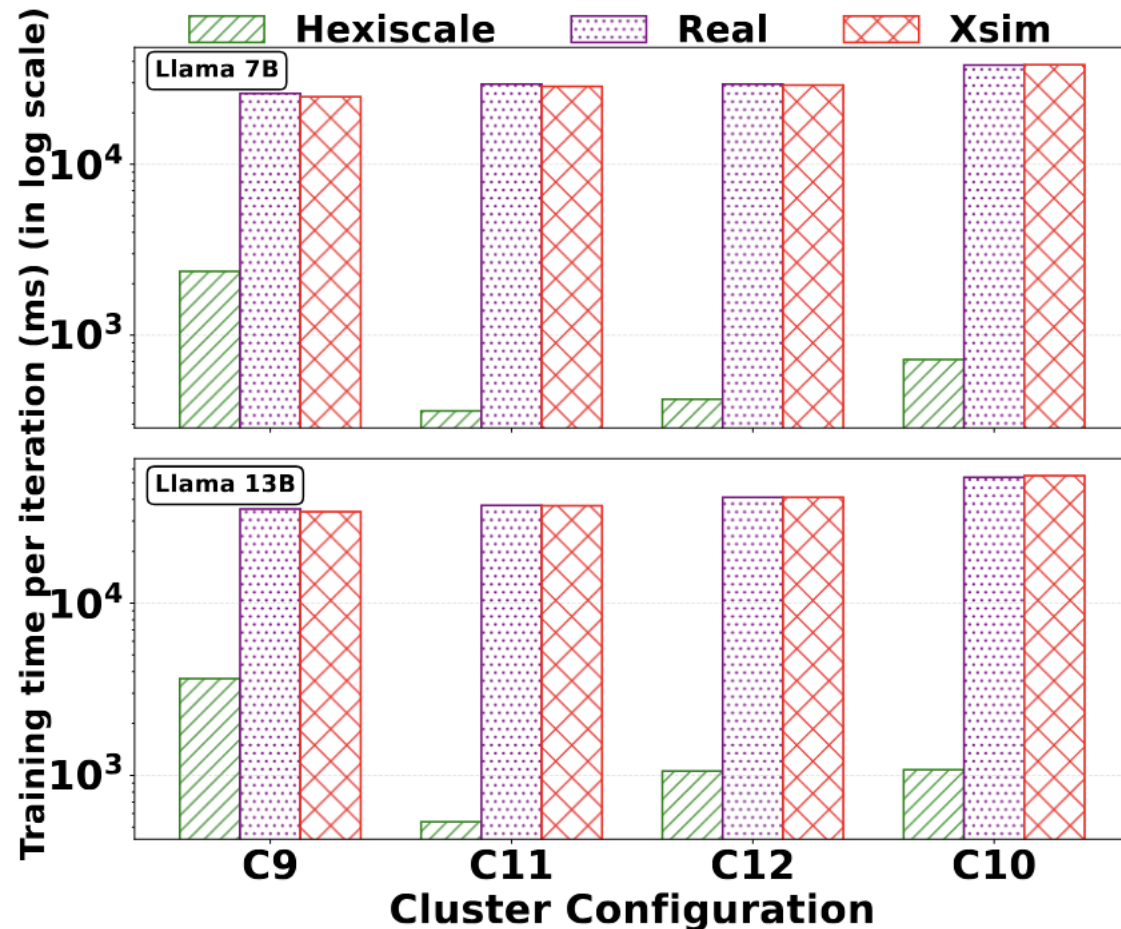


Simulation Setup



Note: Heterogeneous GPU configurations are mentioned in every experiment.

Accuracy : How accurately does Xsim perform on heterogeneous deployment with ground truth and analytical simulator?

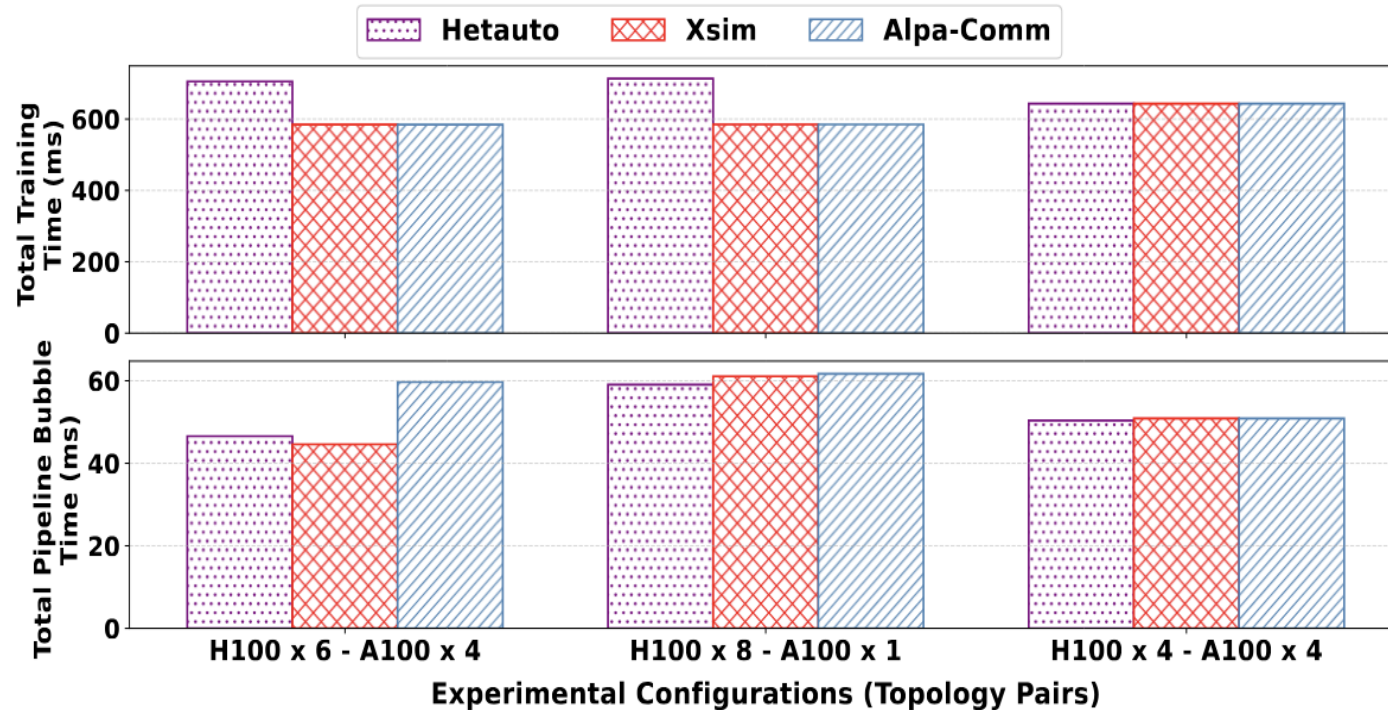


Hexiscale underestimates heterogeneous training time by 90%.

Xsim achieves near-real accuracy (<5% error) by modeling full-stack communication effects.

Config	Compute Cluster	Model Distribution
C9	1 X A100, 1 X H100	DP
C10	2 X A100, 2 X H100	A(2 X DP)-DP-H(2 X DP)
C11	2 X A100, 2 X H100	A(2 X TP)-DP-H(2 X TP)
C12	2 X A100, 2 X H100	A(2 X PP)-DP-H(2 X PP)

Xsim performance with SOTA resharding solutions:



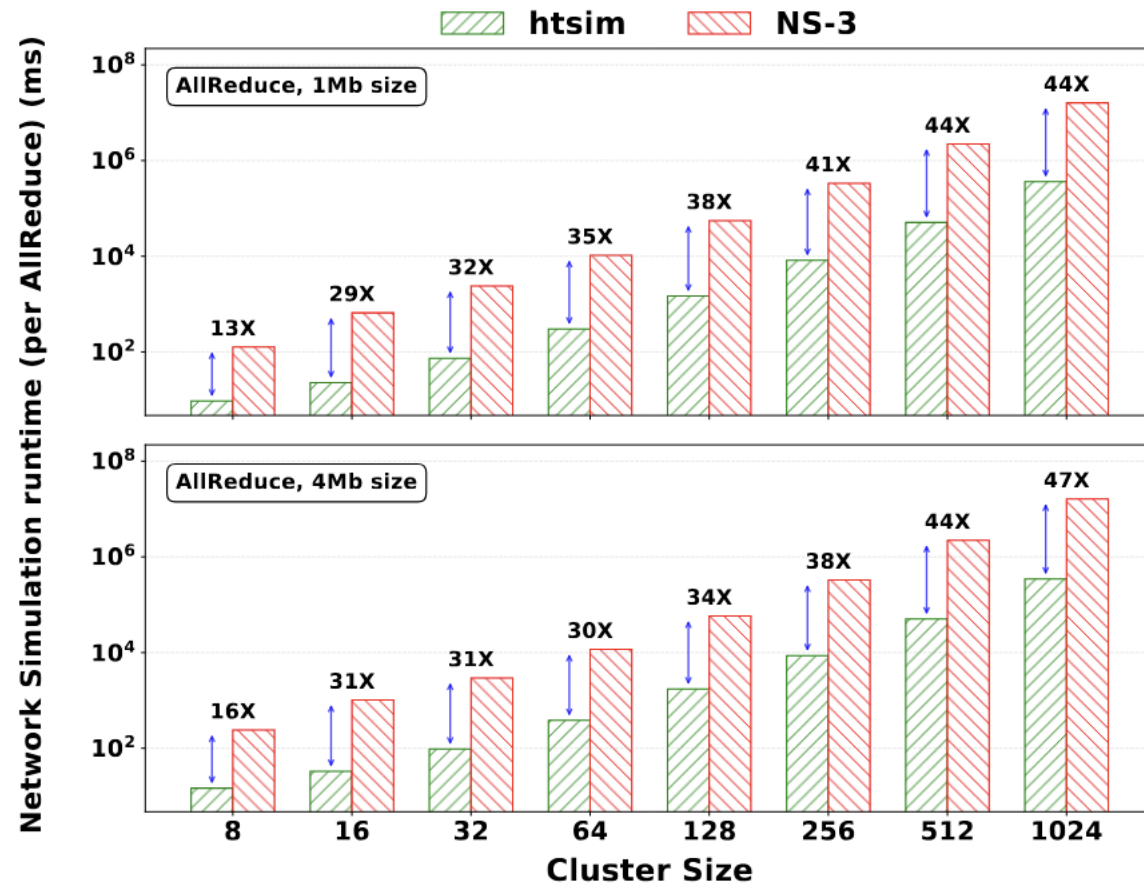
Training time :

- **Hetauto:** Higher in asymmetric setups due to a multi-phase, hierarchical workflow.
- **AlpaComm & Xsim:** Lower because they use direct, point-to-point communication to skip extra aggregation phases.

Pipeline Bubble time:

- **Xsim & Hetauto:** Lower because they partition tensors into balanced units, minimizing idle time.
- **AlpaComm:** Higher due to non-uniform slicing that causes load imbalance and communication skew.

Scalability: How do htsim and NS-3 affect the simulation time in Xsim ?



htsim is 13 – 44× faster than NS-3 across cluster sizes,

htsim → large-scale studies
NS-3 → detailed, high-fidelity network behaviour.

Conclusion and Future work

Conclusion:

- Designed simulator for heterogeneous hybrid-parallel training
- Extensible abstractions + NS-3/htsim backends (fidelity vs. scalability)
- Unified evaluation platform for heterogeneous clusters

Future work:

- *Xsim* to co-evolving models, parallelization strategies & support for inference workload.

Thank You

Questions?